Perception-Oriented UAV Image Dehazing Based on Super-Pixel Scene Prior

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***Abstract*— Current unmanned aerial vehicle (UAV) image defogging techniques often result in images with chromatic aberrations, color distortions, and increased noise due to flight variables and environmental factors, impacting downstream mission objectives. This article presents a novel UAV image dehazing framework designed to enhance perceptual tasks in foggy conditions. Moving beyond traditional pixel-level dehazing, our approach utilizes a super-pixel scene prior (SPSP) method, improving the UAV defogging process. By shifting the dehazing operation from RGB to Lab color space using SPSP, we minimize chromatic confusion and pinpoint reliable defogging areas, especially in the *L* channel. To address light inconsistency challenges during defogging, we introduce a new guided filtering algorithm that leverages simple linear iterative clustering (SLIC). This algorithm utilizes super-pixel clusters instead of large guided windows, preserving crucial information and boosting efficiency. SPSP also guides the SLIC algorithm in Lab space, facilitating faster defogging. Our framework incorporates a quantitative analysis of super-pixel segmentation and target detection, utilizing a feedback loop with the alternating direction multiplier method (ADMM) to optimize perception and defogging concurrently, thus enhancing UAV visual capabilities in fog. Our UAV image dehazing technique outperforms existing methods, as evidenced by quantitative and qualitative assessments, effectively eliminating haze from UAV images and significantly improving perceptual processing in foggy conditions.**

***Index Terms*— Perception-oriented defogging, super-pixel scene prior (SPSP), uneven illumination correction, unmanned aerial vehicle (UAV) image haze removal.**

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I. Introduction

UNMANNED aerial vehicles (UAVs) equipped with camera systems play a pivotal role in various fields, including urban surveillance, infrastructure inspection, and agricultural management, where they perform crucial perceptual tasks. However, UAV-captured images are often compromised by haze or fog due to environmental factors like climate change and pollution, adversely affecting image quality. The scattering of light by suspended particles such as water vapor and dust reduces clarity and contrast, resulting in color degradation and increased blurriness. These issues not only diminish the images' detail and sharpness but also hinder the effectiveness of UAV-based surveillance and analysis. Therefore, developing effective defogging techniques is essential to preprocess these affected images, ensuring the accurate functionality of UAVs' perception systems.

Traditional image processing techniques [1], [2], [3], [4] aimed at enhancing image clarity by adjusting contrast, brightness, and saturation have been found inadequate for addressing image fogging effectively. These basic adjustments fall short of comprehensively defogging images. Consequently, researchers have turned to a priori knowledge from physical models, employing various analytical and statistical methods [5], [6], [7], [8], [9], [10] to enhance defogging performance. In recent years, the advancement of deep learning-based image and video processing technologies has not only expanded the possibilities for clarity enhancement but also brought significant progress to image dehazing methods [11], [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], including parametric generation and direct synthesis approaches. However, these advancements often overlook the impact of dehazing on subsequent visual tasks.

The omission of co-optimization in image enhancement techniques may lead to minimal or negative impacts on the perception essential for advanced visual tasks. Research suggests that without incorporating co-evolutionary methods, traditional image enhancement strategies, including dehazing and deblurring, may not only fail to enhance but could also degrade performance in complex tasks such as image classification, semantic segmentation, and object detection, especially in challenging conditions like low light or rain [23], [24], [25], [26], [27], [28], [29], [30]. This highlights the crucial importance of co-optimization in image processing to significantly improve the perceptual quality necessary for intricate visual tasks.

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In this study, our focus is on developing image-defogging techniques specifically designed for perceptual tasks, aiming to achieve not only effective defogging but also enhanced perceptual accuracy in UAV imagery. Our approach highlights the essential elements of the defogging process, with a special emphasis on enhancing texture features to improve overall perception. By adopting a co-optimization strategy, we aim to progressively enhance the performance of UAV-related applications.

Our methodology comprises two interconnected components. Initially, we select a representative image from a collection of UAV images and transition it to the Lab color space for processing using the super-pixel scene prior (SPSP) method. We employ an advanced homomorphic filtering algorithm for initial defogging, emphasizing the enhancement of white regions. Moreover, we introduce a simple linear iterative clustering (SLIC)-based guided filtering technique that not only accelerates the defogging process, facilitating real-time UAV applications but also mitigates uneven lighting issues, thereby providing an initial reduction in fog.

The second component focuses on improving perceptual aspects. We extract detailed perceptual quantization features both before and after defogging. Using the alternating direction multiplier method (ADMM), we address two convex optimization problems to refine atmospheric light intensity parameters, essential for precise defogging. These optimized parameters are subsequently applied to a sequence of images, completing the defogging process and enhancing the UAV system's perceptual functionality. Through this holistic approach, we aim to contribute to UAV image processing by offering a solution that optimizes defogging efficiency and perceptual clarity without depending on extensive deep-learning resources.

Experimental results unequivocally demonstrate the exceptional performance of our proposed SPSP, establishing a new benchmark in the field. In summary, this article's contributions can be outlined in four key aspects as follows.

1) We present a comprehensive framework to employ a co-optimization approach to address perception challenges in foggy environments, thereby offering valuable assistance to drones in executing a wide range of aerial tasks.
2) We reveal SPSP to precise fog area estimation, upon SPSP, can easily swift image at the super-pixel level and transition to the Lab color space.
3) To rectify the issue of uneven lighting and ensure real-time processing, we enhanced the homomorphic filtering algorithm and refined the guided filtering algorithm by a SLIC-based clustering window.
4) Grounded in the co-optimization principle, we systematically extracted synthesized parameters. Through ADMM, the optimal defogging configuration be attained to ultimately enhance the perceptual efficacy during drone missions.

The remainder of this article is structured as follows. In Section II, we provide a comprehensive review of relevant prior work. Section III presents a detailed exposition of our a priori theory for super-pixel scenes, delves into the methodology for correcting inhomogeneous illumination—including both homomorphic filtering and SLIC optimization-oriented filtering—and introduces perception-oriented defogging optimization algorithms aimed at enhancing the practical utility of defogging. Section IV showcases extensive experiments and evaluations of various UAV defogging methods. In Section V discusses the limitations and outlines avenues for future research, and offers the paper's concluding remarks.

II. Related Work

Image dehazing algorithms are designed to remove distortions from hazy images, mitigating the adverse effects of fog and similar obstructive elements to enhance visual clarity. Contemporary image-defogging algorithms fall into three principal categories: image enhancement, physical modeling, and deep learning-based approaches.

A. Image Enhancement to Remove Haze

Image enhancement methods aim to augment image details and contrast, making the image sharper and more discernible across various applications [31], [32]. Key techniques in this domain include the retinex algorithm [1], histogram equalization [2], and homomorphic filtering [3]. These strategies exploit elements like reflection components, pixel distribution, and frequency domain characteristics to improve image quality. An advancement in this area is Gibson and Nguyen's [4] development of a joint contrast enhancement technique to improve fog removal algorithms. However, while image enhancement is crucial for initial fog removal, it may not fully address the underlying mechanisms of fog formation, potentially leading to over-enhancement and loss of image detail.

B. Physical Model-Based Defogging Algorithms

Physical model-based defogging algorithms utilize atmospheric scattering principles to counteract hazy image formation and restore clarity. A prominent example is the dark channel prior algorithm [5], which identifies relationships between haze-free images and atmospheric scattering parameters, leveraging a large dataset of haze-free images. Renowned for its low computational complexity and high effectiveness, this algorithm has catalyzed the creation of various enhanced methods that extend its foundational principles.

Peng et al. [6] employed depth-dependent color variations to estimate ambient light and ascertain scene transmission parameters within the dark channel prior framework, analyzing the differences between observed intensity and ambient light.

The physical model-based defogging algorithm, rooted in essential principles of image degradation, constrains the imaging model using prior knowledge or a guided learning approach to estimate crucial imaging parameters. These parameters are then incorporated into the model to reverse-engineer a clear, haze-free scene.

Wang et al. [33] assumed a linear relationship between the minimum channel of hazy and haze-free images and proposed a fast dehazing algorithm to balance computational speed and dehazing effectiveness.

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He et al. [7] enhanced the initial transmission map's local consistency by introducing an additional bounding prior. Yuan and Huang [8] used haze-free images as external references for dehazing, applying a scene prior that included structure and high-frequency details, alongside internal constraints and regularization to estimate the transmission map. Li et al. [34] presented a dark direct attenuation prior with a haze line averaging method that uses weighted bootstrap image filters to reduce morphological artifacts and maintain fine structure. Wang et al. [35] proposed a haze line prior for accurate haze feature identification, ensuring quality radiance recovery and preserving structure and color in nighttime images. Ju et al. [36] developed a global strategy using visual metrics and a gamma-corrected prior for scene albedo recovery. Wei et al. [37] introduced a nonhomogeneous dehazing network using an artificial scene prior and 2-D graph inference for propagating the prior between nodes. Zhu et al. [38] created a color attenuation prior to estimating scene depth in hazy images. Guo et al. [9] devised a method to estimate haze thickness, atmospheric light, and transmission using an elliptical boundary prior to pixel clustering in spectral space. Finally, Bi et al. [10] utilized a low-rank and sparse prior to decomposing hazy images' dark channel, separating direct attenuation and atmospheric veil based on atmospheric scattering models.

This method is highly specialized and consistently yields realistic dehazing results for a wide range of images.

C. Deep Learning in Image Dehaze

Deep learning-based dehazing algorithms leverage convolutional neural networks (CNNs) to create end-to-end models that directly derive clear images from hazy ones. These models mainly follow two approaches: the first uses CNNs to derive parameters of the atmospheric scattering model for inverse haze removal [11], [12], [13], [39], while the second employs generative adversarial networks (GANs) to directly synthesize haze-free images from hazy inputs [40], [41], [42], [43]. Although these methods show impressive results on synthetic data, their application to real-world images is often constrained by the scarcity of real-world hazy samples in training datasets.

Zhao et al. [44] combined a priori-based and deep-learning approaches, dividing the dehazing task into visibility restoration and realism enhancement, using adversarial learning with unpaired images to refine the results. Dudhane and Murala [45] developed a CNN architecture to estimate transmission parameters in RGB and YCbCr color spaces, improving robustness for tasks like object detection. Lin et al. [46] introduced a framework using sequential dual-attention deep networks, employing a two-stage learning strategy that first captures fine details, and then thoroughly removes haze.

Yeh et al. [47] proposed a deep-learning architecture leveraging multiscale residual learning and image decomposition to separate hazy images into base and detail components, reducing color distortions in the dehazed output. Li et al. [48] implemented an adversarial framework where a pair of neural networks compete in an end-to-end defogging process, enhancing the model's ability to detect defogging artifacts and manage spatial haze variations. Guo et al. [49] developed a multitasking algorithm for simultaneous raindrop and haze removal, incorporating advanced atmospheric light estimation, transmission map modifications, a GAN, and an optimized visual attention network. Li et al. [50] combined model-based and data-driven methods in a two-scale GAN for haze removal, utilizing transmission maps and atmospheric light estimations for effective dehazing.

Sun et al. [15] introduced PSMB-Net, incorporating a partial Siamese framework (PSF) and a multiscale bi-codec information fusion (MBIF) module, leveraging deep learning for multiscale image analysis and dual codec processing for haze removal. Guo et al. [16] presented a method to synthesize realistic hazy multispectral images and RSDehazeNet, an end-to-end network employing local and global residual learning and channel attention for multispectral image dehazing. Zheng et al. [17] proposed Dehaze-AGGAN, an attention-guided GAN for remote sensing dehazing, using unpaired data and incorporating total variation and cycle consistency losses to enhance edge quality and noise reduction. Chen et al. [18] developed H^2RL-Net, a hybrid high-resolution network for detailed dehazing in remote sensing, addressing nonuniform haze and spatial detail restoration. Li and Chen's [19] FCTF-Net, using a coarse-to-fine attentive process, excels in dehazing synthetic and real-world images. Jiang et al. [20] introduced KFA-Net, designed for dehazing under nonuniform conditions, utilizing an asymmetric feature cascade approach to handle varied haze characteristics in remote sensing images.

In the realm of artificial intelligence, utilizing neural networks for image dehazing has gained prominence. Nonetheless, these deep-learning models require extensive datasets of clear and hazy image pairs for training, which are difficult to obtain naturally. As a result, synthetic methods are often used to create training datasets, but this can lead to training samples that are either incomplete or lack realism.

D. Defogging Algorithms in UAV Images

UAV-captured images often experience reduced visibility and color saturation due to haze, affecting the perceptual functions of intelligent systems. Typically captured from altitudes above 500 m and at downward angles, hazy UAV images display unique traits, such as limited transmission map variations, which can lead to lost detail. In addition, the generally low intensity of UAV images may cause excessive energy reduction and color shifts during defogging. While many still-image defogging algorithms are effective for standard outdoor scenes, their application to UAV aerial video frames can compromise spatiotemporal consistency, causing artifacts like flickering and color changes in the resulting video. Moreover, the high computational demands of these algorithms limit their real-time application. Currently, UAV image-defogging techniques predominantly utilize physical model-based and deep-learning methods, with fewer approaches relying solely on image enhancement.

1) Physical Model Category: In the physical model domain, atmospheric scattering models, a priori information, and UAV-specific data are predominantly used for defogging.

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Wierzbicki et al. [51] developed a defogging technique for UAV-captured images, introducing an atmospheric calibration method based on scattering models. Gao et al. [52] proposed a new atmospheric scattering model for UAV image defogging, incorporating an adaptive variable strategy to enhance atmospheric light accuracy and reduce time complexity. Zhang et al. [53] created a rapid defogging algorithm for UAV images using the dark channel prior, effectively overcoming issues like color distortion and inefficiency, thereby enabling effective defogging, even in images containing sky regions.

Huang et al. [54] developed a defogging algorithm for UAV reconnaissance images using a hierarchical scattering model, incorporating UAV metadata like imaging distances and oblique angles. Zhao et al. [55] improved the dark-channel prior method with a bright region-filling technique to minimize color distortion and improve spatial coherence in UAV image defogging. Huang et al. [56] introduced a defogging algorithm for UAV images based on a variational function and a color-attentive prior, segmenting images into sky and dark regions to accurately resolve atmospheric light values using dark region pixels.

Zhang et al. [57] introduced a dehazing method for UAV remote sensing images, featuring a double-scale transmission optimization strategy to mitigate texture detail loss and color distortion, especially in highlighted areas. This method incorporates a saliency-guided two-scale transmission correction to improve transmission estimation accuracy, essential for maintaining image quality. In addition, a dehaze-driven frequency domain saliency model is employed to enhance the detection and correction of highlighted regions in hazy UAV images, thereby refining the dehazing process.

2) Deep-Learning Category: Within the deep-learning category, prevalent techniques involve the utilization of attention mechanisms, CNNs, dual scaling, and parallel learning.

Wang et al. [58] developed an attention-based feedback-defogging network adept at extracting detailed texture information from UAV remote sensing images, revealing true ground details obscured by haze. Li et al. [59] proposed a CNN-based end-to-end image denoising method that effectively counters haze-induced clarity degradation in high-resolution UAV images. Wu et al. [60] crafted an end-to-end network for UAV image denoising, introducing a loss function with a structural similarity index to maintain essential structural details, thereby enhancing the perceptual quality of dehazed images.

Zheng and Zhang [61] introduced a saliency-guided parallel learning mechanism for UAV image defogging, featuring a parallel defogging module and saliency detection to adeptly process images with sparse objects. Zhang et al. [57] proposed a defogging method for UAV remote sensing images using a two-scale transmission optimization strategy, focusing on accurate transmission map estimation, texture detail preservation, and color accuracy, particularly in highlight regions prone to severe distortion in hazy conditions.

Existing defogging methods within the UAV domain exhibit certain drawbacks, including elevated algorithmic complexity, hampering real-time processing feasibility. Moreover, they often suffer from algorithmic instability and limited applicability. In addition, these methods typically lack adaptive enhancements for postdefogging perceptual tasks, thus failing to fully realize the potential value of the image after the haze removal process.

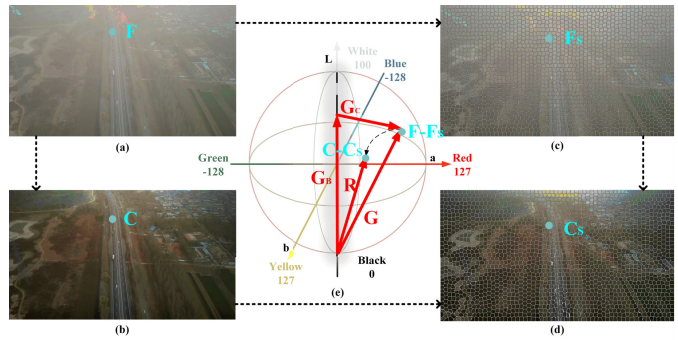

Fig. 1. Schematic of the SPSP: (a) UAV foggy image, (b) UAV dehazed image, (c) super-pixel segmentation result of UAV foggy image in Lab space, (d) super-pixel segmentation result of UAV dehazed image in Lab space, and (e) decomposition result of various types of images in Lab color space.

III. Proposed Method

A. Super-Pixel Scene Prior

UAV image dehazing presents a considerably more complex challenge than dehazing in conventional scenes. To address this complexity, UAV dehazing is divided into two distinct tasks: white area removal and uneven illumination correction. In this context, we introduce a straightforward yet highly effective method called SPSP, which rapidly segregates and reconstructs hazy images into light and dark components. These components are then used to facilitate white area removal and uneven illumination correction in UAV haze scenes, resulting in the generation of high-quality, clear dehazed UAV images.

Drawing inspiration from the gray haze-line prior [35], we introduce a novel concept known as the SPSP. In this approach, the haze primarily affects the brightness component "L" in the Lab color space after segmenting the image into super-pixels, while having a lesser influence on the chromaticity components "a" and "b." We then use these two components to eliminate the haze from the image. To elaborate further, the intrinsic color of haze typically appears as white or gray, distributed along the luminance mid-axis within the Lab color space, as depicted in Fig. 1(e). Mathematically, the gray component of the haze image is identified as the largest common vector across its three channels, a characteristic that can be precisely defined as follows:

$$G_B(x) = (\xi, \xi, \xi)^T, \quad \xi = \min_{C \in \{L,a,b\}} (G_C(x)) \tag{1}$$

where ξ is a constant in the range [0, 1]. Then, we get the corresponding color components

$$G(x) = G_B(x) + G_C(x). \tag{2}$$

To streamline the feature dimension and efficiently remove haze, we project the gray component (G) onto the "L" channel of the Lab color space, utilizing the linear relationship between

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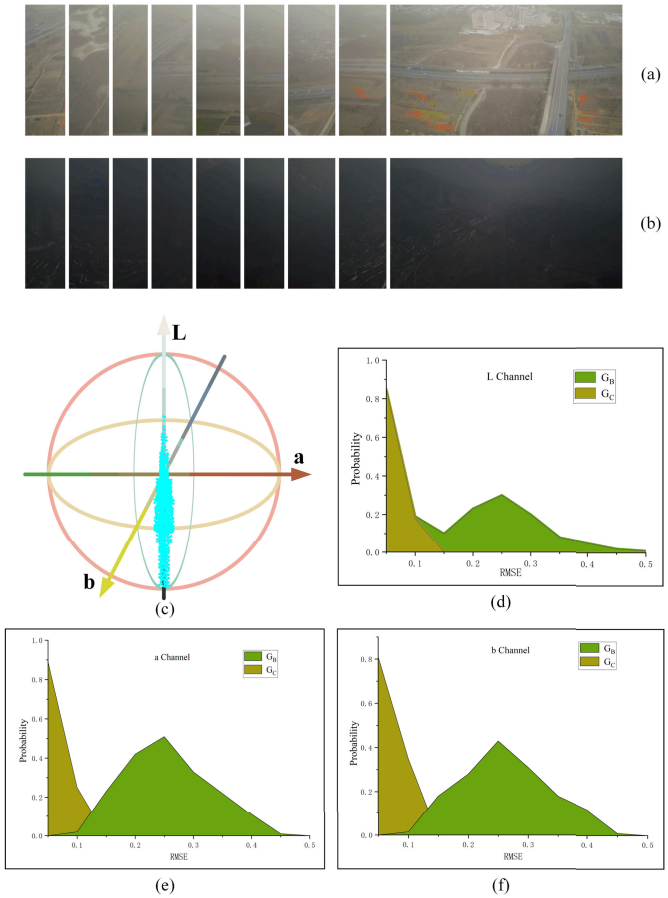

Fig. 2. UAV foggy images and their data statistics: (a) example of UAV foggy image in higher illumination case, (b) example of UAV foggy image in lower illumination case, (c) statistical graph of average fog intensity, (d) RMSE of chromaticity versus luminance components of the L-channel, (e) RMSE of chromaticity versus luminance components of the a-channel, and (f) RMSE of chromaticity versus luminance components of the b-channel.

RGB and Lab color spaces. This process allows us to accurately identify the haze-affected regions. The statistical and theoretical foundations of our proposed method are detailed as follows.

1) Statistical Analysis: To assess the effectiveness of the proposed prior, we conducted a data collection effort using UAVs, amassing a dataset of 3000 real-world images captured under adverse weather conditions, including haze and dust. Fig. 2(a) shows several examples of images taken by UAVs, illustrating the presence of sand and dust in challenging weather conditions. In contrast, Fig. 2(b) features additional examples captured during hazy conditions in the early morning and evening. Analyzing the images in the Lab color space, we computed the mean difference—i.e., the average fog intensity—across all sample pairs, along with the root mean square error (RMSE) for both the color and gray components. As depicted in Fig. 2(c), the mean haze intensity of all sample pairs is notably concentrated along the central axis of the Lab color cube. Moreover, Fig. 3(d)–(f) shows that the RMSE values for the chromaticity components of all channels generally remain below 0.1, whereas the RMSE values for the luminance components exhibit higher variability. This observation suggests that haze primarily impacts

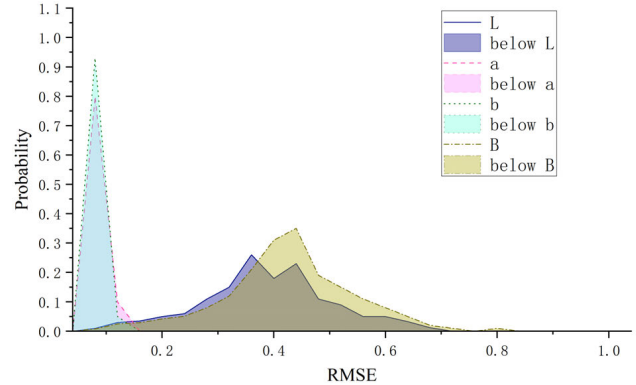

Fig. 3. Statistical analysis of the precise fog area in the Lab color space. The $L/a/b$ is the corresponding color channel, and B is the brightness component of the L channel.

the luminance component while demonstrating less influence on the chrominance component, aligning with the principles underlying our proposed prior approach. Consequently, these comprehensive statistical experiments have provided insights into the distribution of haze features within the 3-D Lab color space, further substantiating the effectiveness of the SPSP method.

Subsequently, we apply the proposed a priori method to identify precise hazy feature regions within the super-pixel segmentation map. To ensure consistency, we normalize the L, a, and b channels of all samples to the [0, 1] range. We then calculate the RMSE for each channel among all sample pairs, including the RMSE for the corresponding gray component in the Lab color space, as presented in Fig. 3(a) and (b). In Fig. 3(b), the RMSE values for the a and b channels generally remain below 0.1, whereas Fig. 3(a) shows more significant variation in RMSE values for the L channel, highlighting the substantial impact of haze on this channel. Moreover, the RMSE values for the luminance components in Fig. 3(a) consistently exceed those of the L channel, indicating that haze more prominently affects the luminance components. Therefore, haze is effectively mapped to the gray component of the L channel, reducing the extent of the haze feature region. This analysis accurately identifies the haze feature regions within the Lab color space, facilitating precise haze removal.

2) Theoretical Analysis: In the presence of atmospheric haze, light interacting with object surfaces is scattered by airborne particles during its propagation to the UAV payload, undergoing continuous attenuation throughout the transmission process. Following this principle, the physical model of atmospheric scattering, which elucidates image degradation in foggy conditions, is expressed as follows:

$$F(x) = D_{\mathrm{eg}} + A_{\mathrm{li}} \tag{3}$$

where $F(x)$ is the original foggy image captured by the UAV load; D_{eg} is the attenuation degradation model, which represents the attenuation degradation process of light propagation in the transmissive suspended particles and can be expressed as

$$D_{\mathrm{eg}} = Q(x)T(x) \tag{4}$$

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where $Q(x)$ is the clear image to be recovered and $T(x)$ is the atmospheric transmission parameter rate.

A_{li} is the atmospheric light model, representing the scattering effect of the overall atmospheric all-optical path, which can be expressed as

$$A_{\text{li}} = A[1 - T(x)] \tag{5}$$

where A is the atmospheric light intensity parameter. With a foggy image $F(x)$ and a clear image $Q(x)$, it can be derived that the foggy entity $H(x)$ can be expressed as

$$H(x) = F(x) - Q(x). \tag{6}$$

As illustrated in Fig. 2(a) and (b), the intrinsic color of haze in the UAV environment's uniform atmosphere tends to be white or gray, resulting in the haze entity $H(x)$ having approximately uniform pixel values across all Lab color space channels. The distribution of $H(x)$ features is predominantly concentrated along the luminance axis, as further affirmed by the statistical findings in Fig. 3, which highlight the significant impact of haze on the luminance component. To effectively mitigate haze features, we project the luminance component onto the L-channel in the Lab color space, enabling precise identification of the haze feature region.

When transforming an image from RGB to Lab space, the channels must first be converted to XYZ space, as detailed in the following equation:

$$\begin{bmatrix} X \\ Y \\ Z \end{bmatrix} = [\theta] \cdot \begin{bmatrix} \mathrm{R} \\ \mathrm{G} \\ \mathrm{B} \end{bmatrix} \tag{7}$$

where $[\theta]$ is a 3×3 matrix

$$[\theta] = \begin{bmatrix} 0.412453 & 0.357580 & 0.180423 \\ 0.212671 & 0.715160 & 0.072169 \\ 0.019334 & 0.119193 & 0.950227 \end{bmatrix}.$$

After that, it is converted from XYZ space to Lab space, and the conversion relation is shown in the following equations:

$$L = 116 * f(Y/Y_n) - 16 \tag{8}$$

$$a = 500 * (f(X/X_n) - f(Y/Y_n)) \tag{9}$$

$$b = 200 * (f(Y/Y_n) - f(Z/Z_n)). \tag{10}$$

In (8)–(10): X_n, Y_n, Z_n-0.950456,1.0,1.088754.

The function $f() - f(t)$

$$= \begin{cases} t^{\frac{1}{3}}, & t > 0.008856 \\ \frac{1}{3}\left(\frac{29}{6}\right)^2 t + \frac{4}{29}, & t \leq 0.008856 \end{cases}$$

which is similar to the gamma function.

According to the above analysis, the atmospheric scattering model can be projected to the L channel of Lab color space, which is formally expressed as

$$\mathcal{F}^L(x) = \mathcal{D}_{\text{eg}}^L + \mathcal{A}_{\text{li}}^L \tag{11}$$

$$\mathcal{D}_{\text{eg}}^L = \mathcal{Q}^L(x)\mathcal{T}^L(x) \tag{12}$$

$$\mathcal{A}_{\text{li}}^L = \mathcal{A}^L\left[1 - \mathcal{T}^L(x)\right] \tag{13}$$

where $\mathcal{F}$, $\mathcal{D}_{\text{eg}}$, $\mathcal{A}_{\text{li}}$, $\mathcal{Q}$, $\mathcal{T}$, and $\mathcal{A}$ are the corresponding Lab vectors of F, D_{eg}, A_{li}, Q, T, and A, respectively. The superscript L represents the L channel in the Lab space.

The foggy entities are mainly distributed on the L channel in the Lab color space. therefore, the following conclusions can be drawn:

$$\triangle\mathcal{H}(x) = \mathcal{F}^L(x) - \mathcal{Q}^L(x). \tag{14}$$

In the L-channel, $\mathcal{F}^L$ and $\mathcal{Q}^L$ can be expressed, respectively, as

$$\mathcal{F}^L(x) = \mathcal{F}_B^L(x) + \mathcal{F}_C^L(x) \tag{15}$$

$$\mathcal{Q}^L(x) = \mathcal{Q}_B^L(x) + \mathcal{Q}_C^L(x). \tag{16}$$

In the above equation, $\mathcal{F}_B^L$ and $\mathcal{Q}_B^L$ are the chromaticity components of $\mathcal{F}$ and $\mathcal{Q}$. Similarly, $\mathcal{F}_B^L$ and $\mathcal{Q}_B^L$ are the chromaticity components of $\mathcal{F}$ and $\mathcal{Q}$, which represent the intrinsic illuminance.

In the presence of hazy weather, our objective is to retain the intrinsic colors of the entire scene both before and after the haze removal process. To achieve this color consistency in hazy scenes, we derive $\mathcal{F}_C^L(x) = \mathcal{Q}_C^L(x)$ By bringing in the above equation, we can get the accurate haze substantial feature region as follows:

$$\triangle\mathcal{H}(x) = \mathcal{F}_B^L(x) - \mathcal{Q}_B^L(x). \tag{17}$$

As a result, the proposed prior theory confirms the accurate projection of haze onto the luminance component within the L channel. Furthermore, when dealing with a haze scene from the UAV perspective, we can derive the chromaticity component $\mathcal{Q}_C = (\mathcal{Q}_C^L, \mathcal{Q}^a, \mathcal{Q}^b)$ and the luminance component $\mathcal{Q}_B = \mathcal{Q}_B^L$ in the Lab color space. These components are instrumental for tasks such as white area removal and inhomogeneous illumination correction.

The atmospheric scattering model for the L channel is written as

$$\mathcal{F}_B^L(x) = \mathcal{Q}_B^L(x)\mathcal{T}^L(x) + \left[\mathcal{A}^L - \mathcal{F}_C^L(x)\right]\left[1 - \mathcal{T}^L(x)\right]. \tag{18}$$

In conclusion, the comprehensive statistical and theoretical analyses presented above confirm the robustness and efficacy of the proposed SPSP method. This prior not only demonstrates that haze predominantly clusters along the bright channel line within the super-pixel scene but also highlights its precise projection onto the luminance component of the L channel in the Lab color space. Building upon this insight, we derive a modified atmospheric scattering model, as described in (18), and subsequently construct a cohesive framework for UAV haze removal within the Lab color space.

B. Haze Correction In Uneven Illumination

To mitigate the impact of uneven illumination and accentuate image details, we employ a frequency domain-based homogeneous filtering algorithm during the initial image processing, which preserves the quality of highlighted areas and concurrently enhances image details.

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1) Improved Homogeneous Filtering Algorithm: The SLIC algorithm [62] is employed to segment pixels based on their similarity in color and distance. This algorithm offers notable advantages, such as high efficiency and the generation of super-pixel blocks with uniform size and regular contours. It is extensively used in image segmentation for UAVs, remote sensing applications, and other fields. The original SLIC super-pixel segmentation algorithm includes the following steps: 1) initialization of cluster centers by dividing the grid according to the image size; 2) adjustment of cluster centers to positions with the minimum gradient in adjacent intervals; 3) determination of the best-matching pixels using a distance measurement formula within the spatially adjacent region around each cluster center; and 4) calculation of the error in the new cluster centers, ensuring connectivity once a certain threshold is met. The distance measure employed is as follows:

$$J = J_{\text{Lab}} + mJ_{xy}/\tau \tag{19}$$

$$J_{\text{Lab}} = \sqrt{(L_j - L_{\hbar})^2 + (a_j - a_{\hbar})^2 + (b_j - b_{\hbar})^2} \tag{20}$$

$$J_{xy} = \sqrt{(x_j - x_{\hbar})^2 - (y_j - y_{\hbar})^2} \tag{21}$$

where J is the distance of each pixel point from the clustering center; J_{Lab} is the color feature distance; (L_j, a_j, b_j) and $(L_{\hbar}, a_{\hbar}, b_{\hbar})$ are the Lab color space coordinates of grid pixel points j and $\hbar$, respectively; J_{xy} is the spatial feature distance; (x_j, y_j) and $(x_{\hbar}, y_{\hbar})$ are the 2-D spatial coordinates of points j and $\hbar$, respectively; m is the spatial distance influence factor; and τ is the distance between seed points.

The homogenous filtering of the image is grounded in the image model, which takes into account incident and reflected light. The image function $g(x, y)$ is represented as the product of the illumination function $i(x, y)$ and the reflection function $r(x, y)$. The image model and its range of values are as follows:

$$g(x, y) = i(x, y) \times r(x, y) \tag{22}$$

$$0 < i(x, y) < \infty, \quad 0 < r(x, y) < 1. \tag{23}$$

By applying a logarithm to (22), performing a Fourier transform in the frequency domain, and selecting an appropriate filtering function to control the right-hand side component, we can derive the filtering result

$$\mathbb{E}(u, v) = \mathbb{S}(u, v) \times \mathbb{G}(u, v) = \mathbb{S}(u, v) \times \mathbb{I}(u, v) + \mathbb{S}(u, v) \times \mathbb{R}(u, v) \tag{24}$$

where $\mathbb{G}(u, v)$, $\mathbb{I}(u, v)$, and $\mathbb{R}(u, v)$ are the frequency domain functions of $g(x, y)$, $i(x, y)$, and $r(x, y)$, respectively; $\mathbb{S}(u, v)$ is the filter function, which is commonly used as Gaussian and Butterworth-type filter function, but it has a large number of parameters and is complicated to calculate.

This article proposes an improved homogeneous filter function

$$\mathbb{S}(u, v) = \alpha[1 + \mathbb{C}(u, v)^{-\beta}]^{-1} \tag{25}$$

where $\mathbb{C}(u, v)$ is the distance from the frequency point (u, v) to the center point of the filter; α and β take values in the range of $0.7 \le \alpha \le 1.4$ and $0.01 \le \beta \le 0.5$, respectively. The improved homogeneous filtering algorithm takes $\alpha = 0.97$, $\beta = 0.28$, the use of homogeneous filtering algorithm on the original image of the light compensation has a certain enhancement effect, but increased a lot of noise, the brightness of the white area has been greatly improved and enhance the overall brightness and clarity, proving the superiority of the improved homogeneous filtering algorithm.

2) Improved SLIC-Based Guided Filtering Algorithm: While the use of the soft keying algorithm for refining $\mathcal{T}(x)$ may introduce higher time complexity, we can enhance both the processing speed and the accuracy of $\mathcal{T}(x)$ calculation by employing the guided filtering algorithm [63]. This can be further optimized and improved by combining it with the downscaled SLIC algorithm.

The guided filter is a smoothing filter designed to preserve edges, effectively smoothing the image background while retaining edge details. Its filter expression is as follows:

$$q = g(I, \varphi, r, \varepsilon) \tag{26}$$

$$q_i = \gamma_s I_i + \delta_s \quad \forall i \in w_s \tag{27}$$

where q denotes the output image; I denotes the guided image; φ denotes the input image; r is the window radius; s is the window pixel; ε is the regularization coefficient; γ_s and δ_s are the window constant coefficients; w_s denotes the filter window, and the constraint equation of w_s is

$$E(\gamma_s, \delta_s) = \sum_{i \in w_s} \left[(\gamma_s I_i + \delta_s - \varphi_i)^2 + \varepsilon \gamma_s^2\right]. \tag{28}$$

The minimization of the deviation between the input and output images leads to the determination of conventional coefficients γ_s and δ_s, expressed as follows:

$$\gamma_s = \frac{\frac{1}{|w|}\sum_{i \in w_s} \varphi_i I_i - \bar{\varphi}_s \mu_s}{\sigma_s^2 + \varepsilon} \tag{29}$$

$$\delta_s = \bar{\varphi}_s - \gamma_s \mu_s \tag{30}$$

where μ_s and σ_s^2 are the mean and variance of the guided map I in window w_s, respectively; $\bar{\varphi}_s$ and $|w|$ are the mean and total number of elements of the input image φ in window w_s, respectively.

The guided filtering window, denoted as w_s, is typically applied within a small localized region, analogous to the super-pixel block near the clustering center in SLIC. In this article, we opt to use the super-pixel block from the SLIC algorithm to compute (29) and (30) for each super-pixel region, effectively reducing the feature dimension of SLIC, as demonstrated in the following equations:

$$\gamma_s = \frac{\frac{1}{\varpi}\sum_{i \in \varpi_s} \varphi_i I_i - \bar{\varphi}_s \mu_s}{\sigma_s^2 + \varepsilon} \tag{31}$$

$$\mathbb{D} = \mathbb{D}_g + \frac{m}{\tau}\mathbb{D}_{xy} \tag{32}$$

$$\mathbb{D}_g = \sqrt{(g_j - g_h)^2} \tag{33}$$

where ϖ denotes the super-pixel block of SLIC; $\mathbb{D}$ is the distance between each pixel point and the clustering center; $\mathbb{D}_g$ is the gray scale feature distance; $\mathbb{D}_{xy}$ is the spatial feature distance; m is the spatial distance influence factor, τ is the distance between the seed points, and g_j and g_h are the

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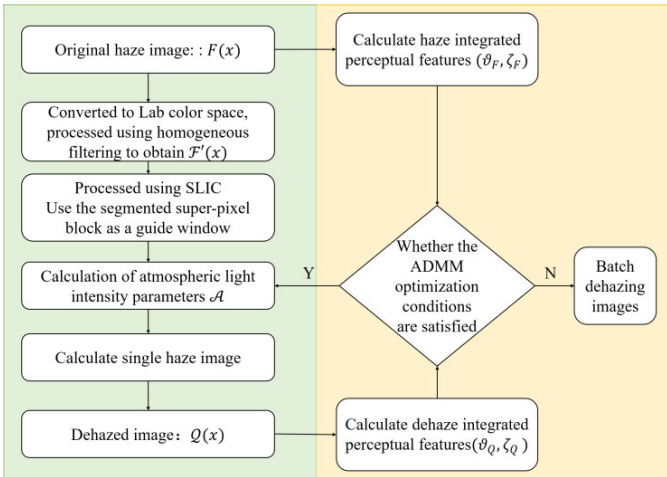

Fig. 4. Overall algorithm flowchart.

gray scale color spatial coordinates of the points j and h, respectively.

Equations (22) and (23) optimize (9)–(11) through iterative calculations that include determining distances from the clustering center to pixel points, searching for similar pixel points near seed points, and computing new clustering centers based on pixel associations. These iterations continue until convergence. The optimized SLIC algorithm, by considering grayscale, horizontal, and vertical spatial features, reduces dimensionality, cuts computational complexity, and minimizes perceptual distortion in the original image. An enhanced SLIC-based guided filtering algorithm is then employed to refine the initial transmittance map, improving computational efficiency and local relationship precision through pixel similarity.

3) Optimized Image Defogging Algorithm: In the generalized defogging algorithm, the atmospheric light intensity parameter is determined based on the maximum pixel value in the fog or the brightest pixel in the image. The presence of intense light irradiation and white objects within the dark channel can significantly impact the estimation, leading to visual distortions or saturation inconsistencies in the restored image. To address this issue, we introduce an adaptive calculation, integrated into the SLIC optimization-based UAV image defogging algorithm, to refine the defogging process. Fig. 4 illustrates the algorithm flow, with the left side showing the flow from Section III-A and the right side from Section III-B.

We begin with the input foggy image $\mathcal{F}(x)$. After converting the color channels, we use the homogeneous filtering algorithm described in Section III-A1 to remove the influence of white areas. Next, the dark channel prior algorithm from Section II-B is applied to compute the dark channel map and the transmittance of the filtered image, $\mathcal{F}'(x)$. The super-pixel segmentation of $\mathcal{F}'(x)$ is then conducted using the SLIC optimization algorithm from Section III-A2, where the generated super-pixel block and super-pixel image serve as guiding windows and maps for filtering. For the calculation of the atmospheric light intensity parameter A, (34)–(36) are employed to allow adaptive adjustment. Finally, the derived parameters are applied in (18) to finalize the recovery of the fog map

$$\hat{\mathcal{F}}^{\text{dark}}(x) = \hat{\mathcal{Q}}^{\text{dark}}(x)\hat{\mathcal{T}}(x) + \mathcal{A}[1 - \hat{T}(x)] \tag{34}$$

$$\max[\hat{\mathcal{F}}^{\text{dark}}(x)] \leq \mathcal{A} \leq \max\left\{\max_{c\in\{L,a,b\}}[\mathcal{F}^c(x)]\right\} \tag{35}$$

$$\mathcal{A} = (1-\lambda)\max[\hat{\mathcal{F}}^{\text{dark}}(x)] + \lambda \max\left\{\max_{c\in\{L,a,b\}}[\mathcal{F}^c(x)]\right\} \tag{36}$$

where $\hat{\mathcal{F}}^{\text{dark}}(x)$, $\hat{\mathcal{Q}}^{\text{dark}}(x)$, and $\hat{\mathcal{T}}(x)$ are the $\mathcal{F}(x)$, $\mathcal{Q}(x)$, and $\mathcal{T}(x)$ processing results of gray scale morphology operation results, respectively. Equation (35) indicates that the atmospheric light intensity parameter $\mathcal{A}$ takes a range of values not greater than the pixel maximum of the original image. Equation (36) in which λ is the adaptive threshold whose value range is [0, 1], and $\lambda = \bar{m}(\hat{\mathcal{F}}^{\text{dark}}(x))$, indicates that taking the mean value of all the elements of $\hat{\mathcal{F}}^{\text{dark}}(x)$, which can reach the function of the adaptive calculation of the correlation of A with the gray scale distribution of the original image.

C. Perception-Oriented Feedback Dehaze Mechanism

In the realm of image defogging, prior research has predominantly relied on conventional machine learning techniques without focusing on enhancing deep visual perception as the primary goal. This limitation has constrained the practical utility of defogging algorithms and diminished their overall significance. By employing a feedback mechanism, we synergistically combine perception and defogging capabilities, achieving holistic enhancement in UAV image defogging. This is realized through the concept of perception-defogging co-evolution, underpinned by the SPSP and the SLIC defogging optimization algorithm.

Conventional UAV image defogging can be achieved using the SPSP and the SLIC defogging optimization algorithm. However, these methods determine the selection of the atmospheric light intensity parameter (A) solely based on image self-adaptation, not directly aligned with the ultimate goal of enhancing perception. To address this, this article introduces a perception-oriented feedback-defogging mechanism. This approach reorients the parameter estimation process in image defogging to align with the broader goal of enhancing perception, offering a novel framework and implementation approach for self-adaptive image defogging.

To enhance the performance of the perception task, we leverage the theoretical insights outlined in Sections III-A and III-B, coupled with the experimental validation presented in this section. In this approach, we utilize SLIC super-pixel segmentation and traditional target detection algorithms to process both predefogging and postdefogging UAV images. These processes are carried out to extract comprehensive representations and define constraints. The selection of appropriate defogging parameters is then achieved through the iterative optimization of individual images. During batch image defogging, we employ the comprehensive representation as a criterion to determine whether to output the defogging results for the current batch or not.

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1) Co-Progression Mechanism of Dehazing and Super-Pixel: The super-pixel segmentation of images containing fog typically yields highly uniform results due to the fog layer present on the image surface. During the segmentation process, the individual super-pixel blocks align neatly with each other. In contrast, the super-pixel segmentation of a defogged image reveals pronounced edge features, making elements such as vehicles, roads, and terrain clearly distinguishable. Consequently, the disparity in edge features within the super-pixel segmented images serves as a valuable feedback indicator. A greater degree of difference signifies a more effective defogging outcome, demonstrating the successful co-evolution of defogging and super-pixel segmentation. Fig. 5 shows the super-pixel segmentation of UAV images in dehaze process.

To facilitate a quantitative comparison between super-pixel segmentation maps before and after defogging, we utilize texture features for information assessment. The texture is not a property of a single-pixel but is manifested as a grayscale feature of the pixel and its surrounding area, often referred to as a local texture feature. A combination of these local texture features contributes to the formation of global texture features.

The local binary pattern (LBP) is an effective operator for texture description. Ojala et al. [64] introduced the LBP as a method for assessing the local contrast of an image. This extended LBP is described by (37), where $\mathbb{Q}, \mathbb{T}$ represents the neighborhood defined by $\mathbb{Q}$ sampling points on a circle with a radius of T, centered at a specified pixel (x_o, y_o)

$$\mathrm{LBP}_{\mathbb{Q},\mathbb{T}(x_o,y_o)} = \sum_{f=0}^{\mathbb{Q}} 2^f s(M_f - M_o) \tag{37}$$

$$s(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases} \tag{38}$$

where M_o is the center pixel point and M_f is the surrounding pixel points of the center pixel point ($f = 0, 1, \ldots, \mathbb{Q}$), each block of super-pixels is considered as an image, and the comparison of textures is performed in terms of super-pixels. The LBP operator produces $2^{\mathbb{Q}}$ different output values corresponding to the $2^{\mathbb{Q}}$ different binary patterns formed by the Q pixels in the neighborhood.

To enhance the comprehensiveness of feature extraction in super-pixel segmentation, we also employed the gray-level co-occurrence matrix (GLCM). GLCM is a commonly used method for describing image textures, involving the exploration of spatial correlation features of gray levels and the spatial distribution relationship between pixels. It possesses strong robustness and adaptability, making it more suitable for texture feature extraction in high-altitude imagery. To meet the needs of different data change detections, four statistical measures of GLCM are used to extract texture features, namely, contrast C_{ON}, which measures the clarity of the image and the depth of texture grooves; energy A_{SM}, reflecting the uniformity of image gray level distribution and the coarseness of texture; entropy E_{NT}, describing the complexity of image gray level distribution; and inverse difference moment H_{NN}, calculating the magnitude of each local change in image texture. To comprehensively select the best texture features, the texture information of super-pixel segmentation before and after defogging is extracted using the pixel window size of the super-pixel block ϖ, obtaining four texture features, respectively.

The SLIC algorithm effectively generates super-pixels with relatively regular shapes in fog-affected images, while it aims to align super-pixels with essential image textures in defogged images. Since the LBP algorithm requires texture extraction within a geometrical shape, this study utilizes the outer rectangle of each super-pixel for such extraction. This outer region's computation is based on the grayscale value of the super-pixel's seed point. Within each super-pixel, individual pixel points are compared in grayscale value across eight different directions. If the grayscale value of a pixel in any direction is greater than that of the central pixel, it is assigned a value of 1; otherwise, it receives a 0.

Each pixel point thus produces an eight-bit binary number, leading to 256 possible feature values, which are subsequently converted into a decimal value to represent the pixel point's LBP feature value. The formula for calculating the superpixel texture features is as follows:

$$s(x) = \begin{cases} 1, & x \geq 0 \\ 0, & x < 0 \end{cases} \tag{39}$$

$$\vartheta = \frac{1}{2\mathbb{W}} \sum_{i \in N} \left(\frac{\mathrm{LBP}_i}{N} + \frac{C_{\mathrm{ON}} + A_{\mathrm{SM}} + E_{\mathrm{NT}} + H_{\mathrm{NN}}}{4N} \right) \tag{40}$$

where N is the number of pixels in the super-pixel block ϖ, LBP_i is the LBP eigenvalue of each pixel, $\mathbb{W}$ is the number of super-pixel blocks ϖ in the whole image, and ϑ is the super-pixel texture feature of the image.

2) Co-Evolutionary Mechanism of Dehazing and Detection: Fog often obscures details in images, affecting the visibility of vehicles, buildings, towers, and other targets, which can result in blurred characteristics and complicate conventional target detection. These complications may include target omissions, false detections, and low-confidence target identifications. However, after defogging, the image is transformed: the fog layer is significantly reduced, leading to clearer and more vibrant textures, and distinct features for various targets in the image. Consequently, target detection on the defogged image yields much more accurate and reliable results compared to the foggy image. Therefore, target detection performance metrics act as essential feedback for fine-tuning defogging parameters, fostering a co-evolution of defogging and target detection that improves overall image quality. Fig. 6 shows the target detection results of UAV images in dehaze process.

In order to conduct a quantitative comparison of the conventional target detection results before and after defogging, the original YOLOv7 algorithm is employed. We utilize a dataset comprising 1000 fog-containing images and 1000 defogged images for training. This dataset is based on the VisDrone dataset. The trained model is subsequently applied to perform target detection on UAV images both before and after defogging. The model's performance is evaluated based on several key metrics, including precision (P), recall (R), confidence (Conf), recall rate (RR), and a weighted summed average score (F). In addition, we measure the detection time (T) for these assessments. The specific definitions for these five metrics are as follows.

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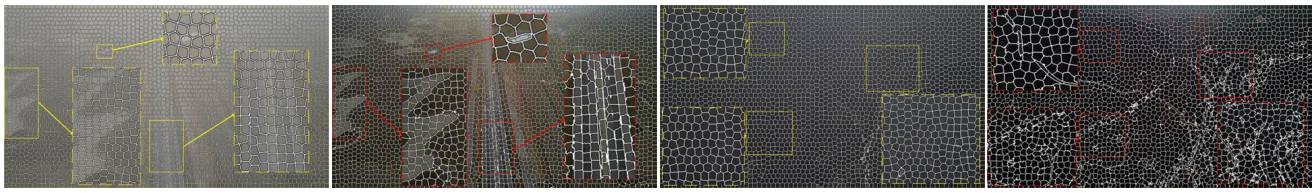

(a) (b) (c) (d)

Fig. 5. Super-pixel segmentation of UAV images in dehaze process: (a) super-pixel segmentation of UAV fog image in high illumination, (b) super-pixel segmentation of UAV defog image in high illumination, (c) super-pixel segmentation of UAV fog image in low illumination, and (d) super-pixel segmentation of UAV defog image in low illumination.

(a) (b) (c) (d)

Fig. 6. Target detection results of UAV images in dehaze process: (a) target detection results of UAV fog image in high illumination, (b) target detection results of UAV defog image in high illumination, (c) target detection results of UAV fog image in low illumination, and (d) target detection results of UAV defog image in low illumination.

TABLE I
CLASSIFICATION OF TARGET DETECTION SITUATIONS

	Positive samples (containing targets)	Negative sample (without target)
Target detected	a	b
No target detected	c	d

The target detection results are presented in Table I, with the following notations: a represents the number of detected targets, b is the count of false detections, c indicates the number of missed detections, and d corresponds to the number of null detections.

Then, P is the accuracy rate and the formula is shown in (41). R is the RR and the formula is shown in (42)

$$P = \frac{\text{Number of positive samples correctly detected}}{\text{Number of positive samples detected}} = \frac{a}{a+b} \tag{41}$$

$$R = \frac{\text{Number of positive samples correctly detected}}{\text{Number of positive samples}} = \frac{a}{a+c}. \tag{42}$$

There exists an inverse relationship between accuracy and recall. If the detection model aims to minimize missed detections, an increase in recall may come at the cost of reduced accuracy, and conversely, prioritizing low false detections might result in a reduction in recall. The ideal objective for the detection model is to achieve high values for both accuracy and recall. To quantify this, the harmonic mean of these two metrics, known as the $F1$-score, is computed, and a higher $F1$-score indicates a better overall performance. The formula for calculating the $F1$-score is presented in the following equation:

$$\mathbb{F} = \frac{(1+\eta^2) * P * R}{\eta^2 P + R}. \tag{43}$$

When $\eta = 1$ in the F-score, P and R occupy the same weight, which is the $\mathbb{F}1$ score, and when $\eta = 2$ in the F-score, it favors R more, which is the $\mathbb{F}2$ score. In order to better evaluate the effect of defogging, it is necessary to pay more attention to the RR, and in order to reduce the phenomenon of missed detection, the $\mathbb{F}2$ score is used in this article as the final precision judgment of the model effect.

3) Perceptual Oriented Dehaze Optimization Algorithm: In Sections III-C1 and III-C2, we acquired perceptually oriented representations of super-pixel texture features before and after defogging, denoted as ϑ, and a composite target detection feature, $\mathbb{F}2$, which we'll denote as ζ for clarity. When perceptually oriented defogging optimization is not in effect, the adaptive computation of $\mathcal{A}$, based on the grayscale distribution of the original image, is determined by calculating the mean of all elements in $\hat{F}^{\text{dark}}(x)$. In this section, we formulate the following equation by combining ϑ and ζ in the context of negative feedback:

$$f(\vartheta) = \vartheta_F - \vartheta_Q \tag{44}$$

$$g(\zeta) = \zeta_F - \zeta_Q \tag{45}$$

where ϑ_F represents the super-pixel texture features of fog-containing images, ϑ_Q represents the super-pixel texture features of defogged images, ζ_F represents the target detection integrated features of fog-containing images, and ζ_Q represents the target detection integrated features of defogged images. According to the analysis in Sections III-C1 and III-C2, the super-pixel texture features and target detection integrated features will be improved after defogging, and the larger the improvement, the better the defogging effect, so the following

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objective function can be constructed:

$$\begin{aligned} &\min\ f(\vartheta) + g(\zeta) \\ &\text{s.t. } \vartheta \geq 0; \quad 0 < \zeta < 1;\ \vartheta + \zeta = \epsilon. \end{aligned} \tag{46}$$

In the above equation, ϵ is the introduced free variation vector.

The objective function can be partitioned into two separate components; however, the interconnected variables are constrained linearly, allowing for a solution through the ADMM. ADMM, originally introduced by Gabay and Mercier [65], was subsequently reviewed and validated for its applicability to large-scale distributed convex optimization problems by Boyd et al. [66], garnering widespread recognition.

The solution is carried out by the ADMM algorithm, and the augmented Lagrange function of the objective function is

$$\begin{aligned} L_\lambda(\vartheta, \zeta; z) = {} & f(\vartheta) + g(\zeta) \\ & + \langle z, \vartheta + \zeta - \epsilon \rangle + \frac{\lambda}{2}\|\vartheta + \zeta - \epsilon\|_2^2 \end{aligned} \tag{47}$$

where z is the Lagrange multiplier and $\lambda > 0$ is the penalty parameter.

For the objective function, the ADMM algorithm is as follows:

$$\vartheta^{k+1} = \arg\min_{\vartheta} L_\lambda(\vartheta, \zeta^k, z^k) \tag{48}$$

$$\zeta^{k+1} = \arg\min_{\zeta} L_\lambda(\vartheta^{k+1}, \zeta, z^k) \tag{49}$$

$$z^{k+1} = z^k + \tau\lambda(\vartheta^{k+1} + \zeta^{k+1} - \epsilon). \tag{50}$$

Use τ as the step size in the multiplicative update, where

$$\tau = \left(0, \frac{1+\sqrt{5}}{2}\right)$$

$$s^{k+1} = \tau\lambda(\zeta^{k+1} - \zeta^k) \tag{51}$$

$$r^{k+1} = \vartheta^{k+1} + \zeta^{k+1} - \epsilon. \tag{52}$$

Equation (51) is pairwise residual, (52) is principal residual, while satisfying the following inequality:

$$\left\|r^k\right\|_2 \leq \varepsilon^{\text{pri}}, \quad \left\|s^k\right\|_2 \leq \varepsilon_{\text{Lab}} \tag{53}$$

where ε is a set accuracy threshold and

$$\varepsilon^{\text{pri}} = \sqrt{p}\varepsilon^{\text{abs}} + \varepsilon^{\text{rel}} \max\left\{\left\|\vartheta^k\right\|_2, \left\|\zeta^k\right\|_2, \|\epsilon\|_2\right\} \tag{54}$$

$$\varepsilon_{\text{Lab}} = \sqrt{n}\varepsilon^{\text{abs}} + \varepsilon^{\text{rel}}\left\|z^k\right\|_2. \tag{55}$$

When the conditions for iterative convergence are met, the optimal defogging parameters for a single image can be obtained, signifying the successful completion of the perceptually oriented defogging optimization algorithm.

IV. Experimental Results and Analyses

This section encompasses subjective, objective, and perceptual evaluations of the defogging results achieved through the proposed method.

Initially, a comparative analysis is performed to assess the effectiveness of the proposed defogging method in contrast to various alternative approaches. Subsequently, the perceptual performance following the defogging process is evaluated. Finally, an ablation study is carried out, providing a comprehensive complexity analysis and showcasing a substantial collection of image enhancement examples for severe weather conditions.

All experiments in this study were carried out on a PC with 64 GB of RAM and a 3.7 GHz CPU using the Visual Studio 2019 (VS2019) platform. Key parameters were configured as follows: $\alpha = 0.97$, $\beta = 0.28$. The threshold ϵ for the perception-directed defogging algorithm was defined as 3×10^{-3}, with a maximum iteration limit of 100. Building on this foundation, we utilized C++ for the algorithm design in this article. The parameters set in the code are as follows: the number of super-pixels is 900, the color similarity threshold is 15, the guided radius is 60, the minimum value radius is 15, and the maximum atmospheric light is 220. Unless explicitly specified, the experiments throughout this research utilized the aforementioned parameter settings.

This study utilized self-collected experimental data. Data acquisition took place at various altitudes and angles using the DJI Mavic 2 Enterprise Advanced UAV in urban, mountain, greensward, and suburban locations within Shijiazhuang City, Hebei Province, China. Data collection occurred in March 2022, December 2022, and March 2023, coinciding with weather conditions that included sandstorms and haze. Images were captured during different times of the day, including early morning, afternoon, and evening. The subjects within the images consisted of a variety of objects, such as power line towers, cars, vans, trucks, minibuses, lodercars, and excavators. All images were captured at a resolution of 1920×1080, resulting in a total dataset of 3000 images.

A. Subjective Assessment

The results of defogging images captured by the UAV in sandy and dusty weather conditions are displayed in Fig. 7. Fig. 7(a) shows the original image taken in a challenging environment. Subsequently, Fig. 7(b)–(h) exhibits the outcomes of applying various defogging algorithms, including the method presented in this article, the HEkaiming algorithm, the D. Wierzbicki algorithm, the H. Yuqing algorithm, the S. Wang algorithm, the Y. Li algorithm, and the R. Zheng algorithm. In Fig. 7(a), it is evident that the top-middle part of the original image has uneven illumination, likely caused by the camera's pitch angle during UAV capture. Fig. 7(c), which shows the result of applying the proposed defogging algorithm, presents a white, patchy area in the top-middle region of the image, with residual uneven coloration. Notably, Fig. 7(d) reveals overall blurriness in the central portion of the image with a distinct defogging effect compared to the other methods. In Fig. 7(e), the algorithm improves the uneven lighting issue in the central region, but overall transparency and brightness remain challenges. Fig. 7(f) displays enhanced contrast in the original image, but inconsistencies in illumination persist in some areas. As shown in Fig. 7(g), color patches are evident in the image after defogging, with pronounced unevenness in illuminated regions. Finally, Fig. 7(h) exhibits a patchy effect and uneven color. In contrast to other algorithms, the results of our proposed method, depicted in Fig. 7(b), demonstrate overall transparency, faithful color restoration, a rich sense

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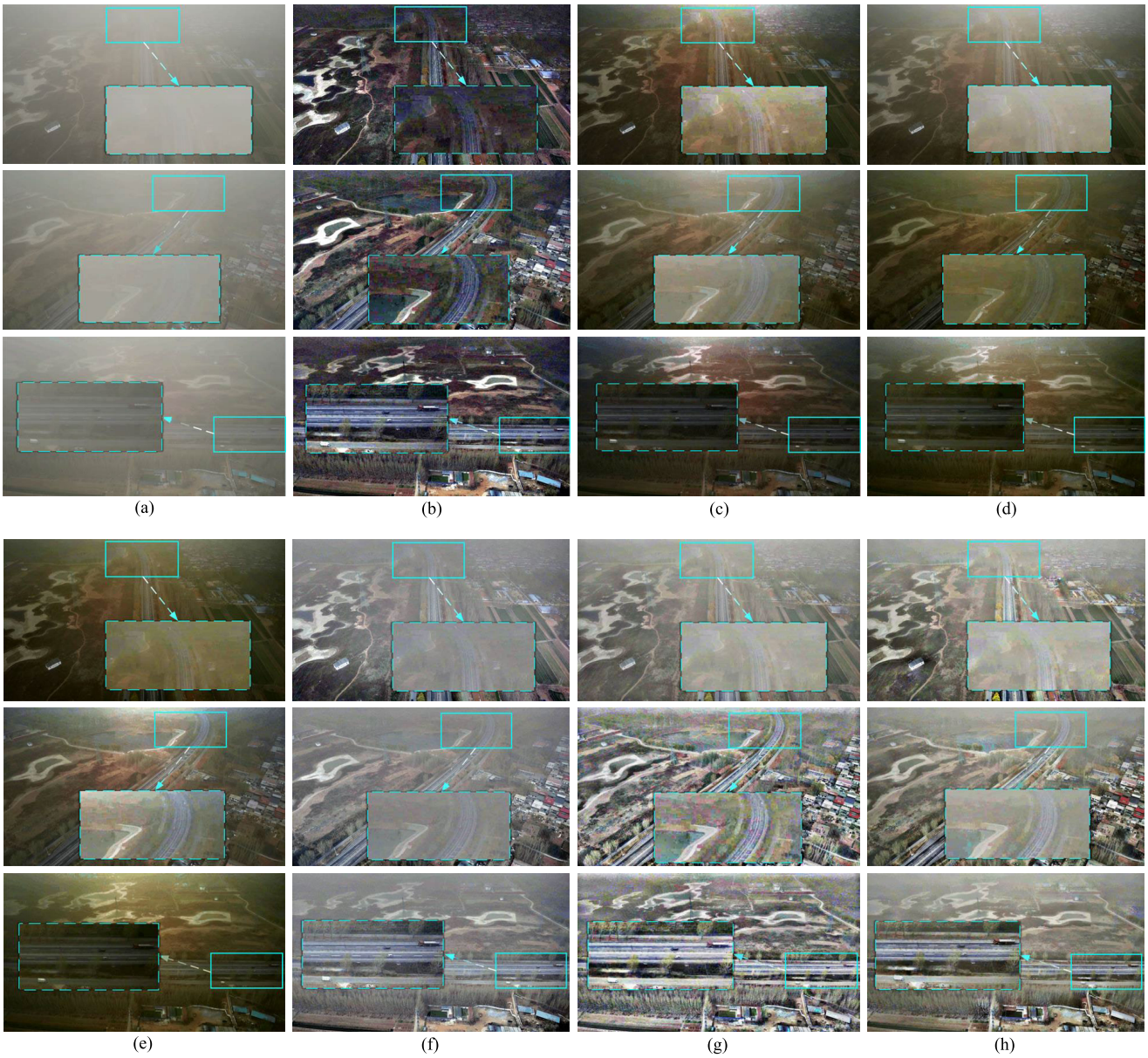

Fig. 7. Results of UAV images in high illumination environments using various dehaze algorithms: (a) original UAV image with fog, (b) algorithms in this article, (c) algorithms in [5], (d) algorithms in [51], (e) algorithms in [54], (f) algorithms in [58], (g) algorithms in [59], and (h) algorithms in [61].

of visual depth, removal of unevenly lit areas, and enhanced object edge details.

To further assess the algorithm's applicability, the results of defogging images captured by the UAV in morning foggy conditions are presented in Fig. 8. Fig. 8(a) shows the original image taken in the foggy morning environment. In this image, the fog obscures details, and the overall light intensity is low, particularly noticeable in the middle-left section of the image, where identifying the highway and trail is quite challenging. Fig. 8(b)–(h) shows the images after applying different defogging algorithms, including the algorithm presented in this article, the HEkaiming algorithm, the D. Wierzbicki algorithm, the H. Yuqing algorithm, the S. Wang algorithm, the Y. Li algorithm, and the R. Zheng algorithm. As seen in Fig. 8(c), the HEkaiming algorithm effectively reduces fog near the camera but diminishes the overall brightness in distant dark areas. Fig. 8(d) shows improved fog removal in the original image, but a white area near the top part of the image is evident due to uneven illumination. Fig. 8(e) exhibits a loss of detail after defogging, along with some noise in distant regions. In Fig. 8(f), the algorithm adapts better to dark scenes but struggles with uneven illumination issues. Fig. 8(g) demonstrates effective defogging in the near-dark scene but significant distortion in the far distance. In Fig. 8(h), the algorithm exhibits decent defogging capabilities in dark scenes but falters in extremely dark areas with uneven color. Compared to other algorithms, our proposed algorithm considers the impact of uneven illumination areas, offering superior fog removal in dark scenes, producing thorough defogging, rendering clear road details, and delivering a visually appealing result.

B. Objective Assessment

The objective evaluation of defogging algorithms relies on quantitative metrics derived from mathematical models,

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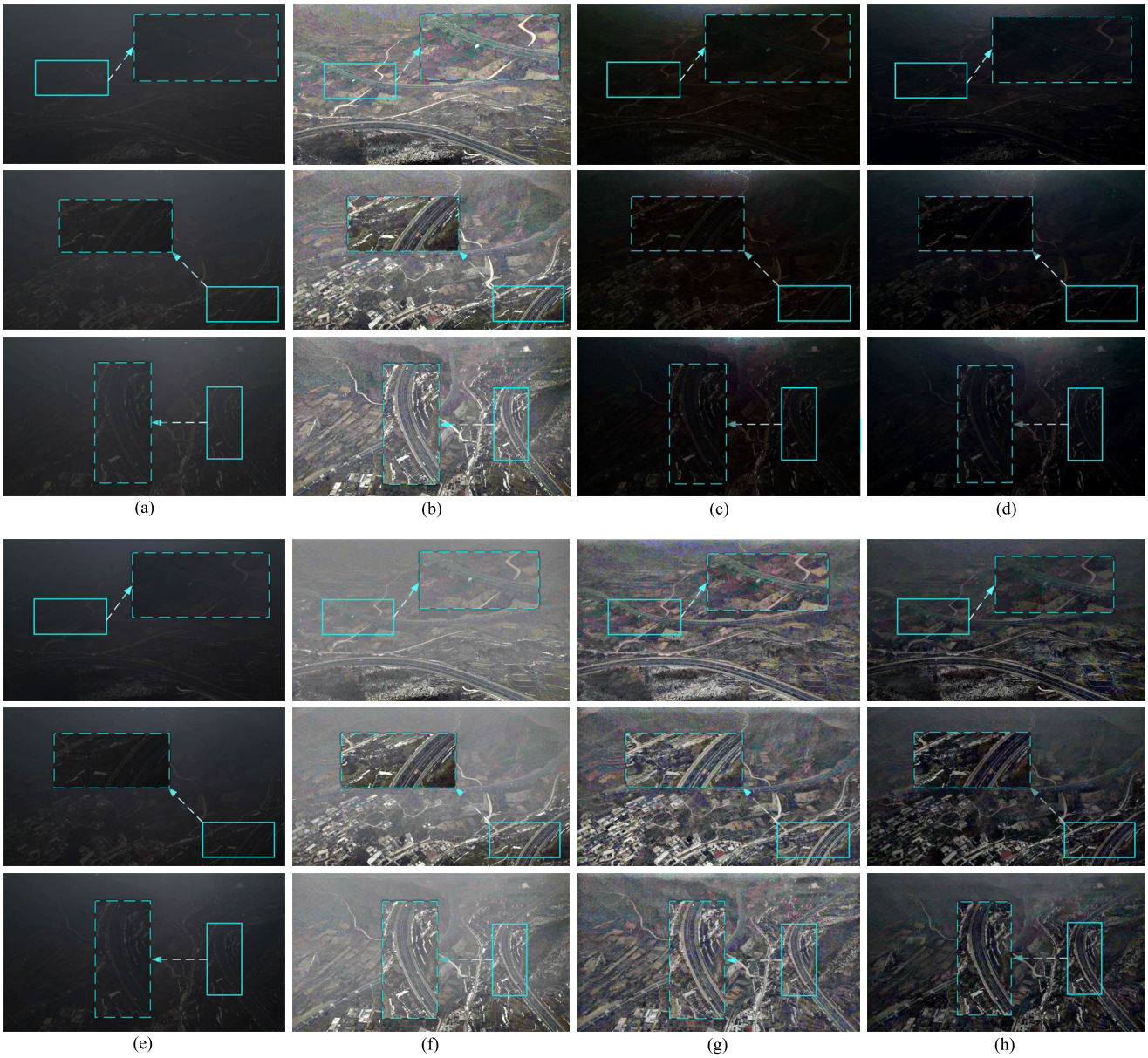

Fig. 8. Results of UAV images in low illumination environments using various dehaze algorithms: (a) original UAV image with fog, (b) algorithms in this article, (c) algorithms in [5], (d) algorithms in [51], (e) algorithms in [54], (f) algorithms in [58], (g) algorithms in [59], and (h) algorithms in [61].

enabling precise measurement of experimental results. In this study, we employ four key metrics, namely, contrast, average gradient, color bias, and information entropy, to assess seven different algorithms. The formulas for each metric are as follows:

$$\Gamma = \sqrt{\frac{1}{n}\left\{\sum \mathcal{L}^2(X) - \frac{1}{n}\left[\sum \mathcal{L}(x)\right]^2\right\}} \tag{56}$$

$$\mathcal{G} = \frac{1}{(h-1)(w-1)}\sqrt{\left(\frac{\partial j(x,y)}{\partial x}\right)^2 + \left(\frac{\partial j(x,y)}{\partial y}\right)^2} \tag{57}$$

$$K = \frac{100\sqrt{\mu_a^2 + \mu_b^2}}{\sqrt{\mu_a^2 + \mu_b^2}\max\left\{\left|\min\left[\left(\sigma_L^2 - T_{\text{th}}, 1\right)\right]\right|, 1\right\}} \tag{58}$$

$$\Lambda = -\sum_{e=0}^{\mathbb{H}-1} p(e)\log_2 p(e) \tag{59}$$

where Γ is the contrast, which indicates the expression ability of image details; $\mathcal{L}(x)$ indicates the luminance of pixel point x; n is the number of pixel points; $\mathcal{G}$ is the average gradient, and h, w, $((\partial j(x,y))/\partial x)$, and $((\partial j(x,y))/\partial y)$ indicate the height, width, horizontal gradient, and vertical gradient of the image, respectively; K is the color deviation, which indicates the degree of color deviation; μ_a and μ_b, respectively, mean value of a and b in Lab space; σ_L is the mean square deviation of L in Lab space; Tth is the deviation threshold; Λ is the information entropy, which indicates the richness of the image information; $\mathbb{H}$ is the number of gray levels; and $p(e)$ is the probability of the distribution of gray levels.

Among the four metrics mentioned above, larger values for Γ, $\mathcal{G}$, λ, and smaller values for K indicate an improved overall image quality after defogging. We calculated and statistically determined the mean value of each index for the images presented in Figs. 4 and 5 using the algorithms described in

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TABLE II
RESULTS OF THE OBJECTIVE ASSESSMENT

Indicators/ Algorithms	Γ	$\mathcal{G}$	K	Λ
this paper	43.825	0.035	1.298	9.591
C	34.513	0.026	1.419	7.682
D	36.362	0.031	1.578	8.043
E	31.639	0.029	1.482	8.541
F	39.541	0.032	1.521	9.379
G	30.776	0.027	1.514	8.102
H	32.614	0.026	1.621	7.651

this article, HE algorithm, and PEI algorithm. The results are summarized in Table II, with the best value for each index highlighted in bold.

As observed in Table II, when compared to other algorithms, our algorithm from this article demonstrates significant improvements in the original image's contrast and effectively reduces the extent of the inconsistent color deviation. In addition, our algorithm records the highest information entropy value, signifying its remarkable capability to remove fog from power inspection UAV images and restore substantial amounts of detailed information within the image.

C. Perception Assessment

To conduct perception-oriented defogging optimization, we evaluated the defogging results from various algorithms and employed YOLOv7, one of the most commonly used target recognition algorithms. Specifically, we defogged 3000 UAV images affected by fog using both the dark channel defogging algorithm and the algorithm presented in this study. The defogged images then underwent target recognition using YOLOv7, with the outcomes analyzed and assessed. Due to equipment limitations, the input image size was scaled down to 800×800 pixels. We employed the stochastic gradient descent (SGD) optimizer with a learning rate of 1.5×10^{-2}, a momentum of 0.85, a weight decay of 6×10^{-4}, and implemented the cosine annealing algorithm for learning rate adjustment. The batch size was set to 8, and the training alternated between 10 training epochs and 1 test epoch for a total of 300 epochs.

In object detection, the configuration of the loss function directly influences the accuracy of the results. The loss function optimizes the positional error between the detected and the ground truth objects, aiming for a prediction box that closely approximates the actual bounding box. Due to inconsistencies in scenes and dataset quality in this study, we use the Wise-IoUv3 version as the loss function for YoloV7 to balance model training results and achieve more uniform detection outcomes. Wise-IoUv3 introduces category weighting to traditional IoU to minimize intercategory differences, assigning weights to categories to adjust IoU overlap calculations, thereby ensuring an unbiased evaluation.

To thoroughly evaluate the target recognition results, we use mean average precision (mAP) as the primary metric. mAP, representing the average of various average precision (AP) scores across the dataset, is derived from the precision–recall (P–R) curve, with mAP50 calculated at an IoU threshold of 0.5. The formula for calculating mAP is as follows:

$$\mathrm{mAP} = \frac{1}{\mathbb{K}} \sum_{\mathbb{K}=1}^{\mathbb{K}} \mathrm{AP}(\mathrm{P}, \mathrm{R}, \mathbb{K}) \tag{60}$$

where $\mathbb{K}$ represents the number of detection target categories, $\mathbb{K} = 9$ in the test of this article, the main targets include person, tricycle, eletower, cars, mixer car, minibus, box truck, huge truck, and tanker car.

The subsequent test images depict the recognition results for the original foggy image, the recognition results for the dark channel defogged image, and the recognition results for the image defogged using the method proposed in this article.

This set of images vividly illustrates the defogging effect and perceptual outcomes for UAV images captured under larger pitch angle conditions. As depicted in Fig. 9(a), the fog significantly impacts the image content, resulting in blurred targets such as vehicles and power line towers. Some targets are barely discernible to the human eye, leading to challenges for target recognition algorithms. Some targets, while recognized by the algorithm, exhibit low confidence levels, introducing uncertainties in subsequent tasks. Fig. 9(b) displays the image after applying the dark channel defogging algorithm. Compared to Fig. 9(a), most of the targets become clearer. However, limitations of the dark channel defogging approach hinder the successful recognition of targets with complex backgrounds, as it lacks the ability to account for perceptual co-evolution factors. Incorrect recognitions or those with low confidence levels persist. Fig. 9(c) reveals the target recognition results achieved with the defogging algorithm proposed in this article. In contrast to Fig. 9(a) and (b), vehicle targets exhibit clear contours, and power line towers are sharply distinguished from the background. The targets with complex backgrounds also gain improved clarity, thanks to the defogging perceptual co-evolution mechanism. The target recognition results have notably enhanced compared to the previous outcomes.

This series of images vividly showcases the impact of defogging on UAV images characterized by a large field of view and dense target areas. As illustrated in Fig. 10(a), the presence of fog obscures the spaces around dense targets, making them nearly imperceptible to both the human eye and target recognition algorithms. In Fig. 10(b), the dark channel defogging algorithm partially mitigates the fog's impact, but some targets remain unrecognized due to insufficient contrast. Moreover, many recognized targets lack high confidence levels. Fig. 10(c) reveals a marked transformation. Compared to the previous images, the fog is virtually eliminated, and target textures are sharp and clear. Notably, the front windshields of some vehicles even show reflective properties, significantly enhancing the target recognition algorithm's effectiveness.

This series of figures showcases UAV images captured under foggy conditions with poor lighting, along with their respective defogged versions and the resulting perceptual outcomes. In Fig. 11(a), the combined effects of fog and inadequate lighting hinder target recognition, leading to low-confidence identifications. In Fig. 11(b), after applying the dark channel defogging algorithm, there is some alleviation

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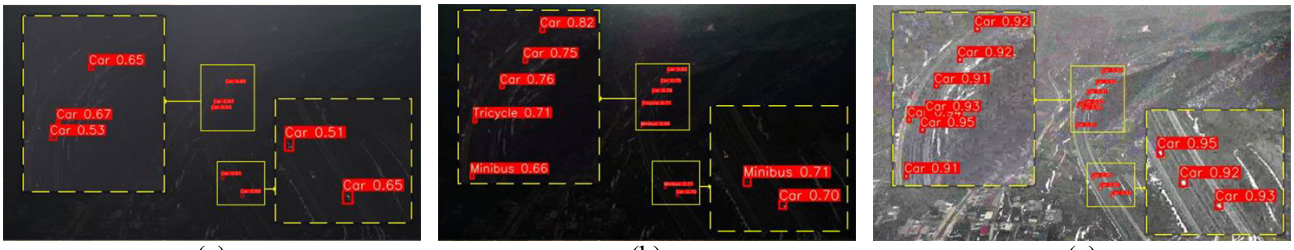
(a) (b) (c)

Fig. 9. Perception results of UAV images during defogging under large pitch angle conditions: (a) perception results of original UAV image with fog, (b) perception results of UAV image using dark channel defogging algorithm, and (c) perception results of UAV image using the algorithms in this article.

(a) (b) (c)

Fig. 10. Perception results of UAV images during defogging under a large field of view and dense target conditions: (a) perception results of original UAV image with fog, (b) perception results of UAV image using dark channel defogging algorithm, and (c) perception results of UAV image using the algorithms in this article.



(a) (b) (c)

Fig. 11. Perception results of UAV images during defogging under low illumination: (a) perception results of original UAV image with fog, (b) perception results of UAV image using dark channel defogging algorithm, and (c) perception results of UAV image using the algorithms in this article.

of the fog's impact. However, the issue of insufficient lighting persists, resulting in many target omissions. Fig. 11(c) displays the perceptual results from the defogging algorithm proposed in this article, which, by leveraging a perceptual co-evolutionary mechanism and incorporating SLIC-optimized filtering improvements, successfully addresses the lighting challenges. The vehicles are clearly restored in terms of contours and colors. Compared to the earlier images, the vehicles now appear white, showcasing the perceptual algorithm's success in achieving comprehensive detection and high confidence levels.

Defogging and perceptual evaluation were conducted on a dataset of 3000 images. We collected statistics on the AP and mAP for the nine categories of objects. The results are presented in the following table.

Looking at the Table III, we can see that the defogging algorithm proposed in this article notably enhances the success rate of object recognition in foggy images and outperforms other defogging algorithms in terms of perceptual improvement.

TABLE III

RESULTS OF THE PERCEPTION ASSESSMENT

Image properties/ Target Type		original figure	Dehaze by [5]	Dehaze by this paper
Person	A	53.5	72.1	91.3
Tricycle	P	62.3	78.3	92.6
EleTower	@	65.4	83.5	95.4
cars	0	66.7	82.9	96.8
MixerCar	.	61.2	76.3	90.3
Minibus	5	63.8	74.9	91.2
BoxTruck	(	56.8	75.6	93.9
HugeTruck	%	64.9	83.4	94.7
Tankcar	)	61.3	81.2	92.8
mAP@0.5(%)		61.77	78.69	93.22

D. Ablation Experiment

To verify the effect of different modules and the speed advantage of the proposed method, ablation experiments were conducted on the dataset in this article. To ensure the fairness of the comparison, all programs of the different methods are run in the same environment and parameters, test image resolution is 1920 × 1080.

In Table IV, our algorithm builds upon the SPSP, investigating the impact of two modification algorithms and one mechanism on the performance of our algorithm. The experimental results in terms of mAP indicate that the perception-oriented feedback dehaze mechanism significantly

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TABLE IV
RESULTS OF ABLATION EXPERIMENT

Improved Homogeneous Filtering Algorithm	**Improved SLIC-based Guided Filtering Algorithm**	**Perception-oriented Feedback Dehaze Mechanism**	mAP@0.5(%)	Γ	$\mathcal{G}$	K	Λ	**Time (ms)**
Addition	Addition	Addition	93.22	43.761	0.034	1.313	9.582	47
Addition	No Addition	No Addition	82.76	37.372	0.025	1.457	8.827	51
No Addition	Addition	No Addition	83.41	37.181	0.026	1.462	9.073	52
No Addition	No Addition	Addition	89.57	33.294	0.021	1.609	7.681	56
Addition	Addition	No Addition	86.29	39.307	0.032	1.351	9.492	48
No Addition	Addition	Addition	91.85	37.492	0.026	1.451	9.108	51
Addition	No Addition	Addition	90.63	37.518	0.025	1.403	8.937	51
No Addition	No Addition	No Addition	81.92	32.154	0.023	1.611	7.739	57

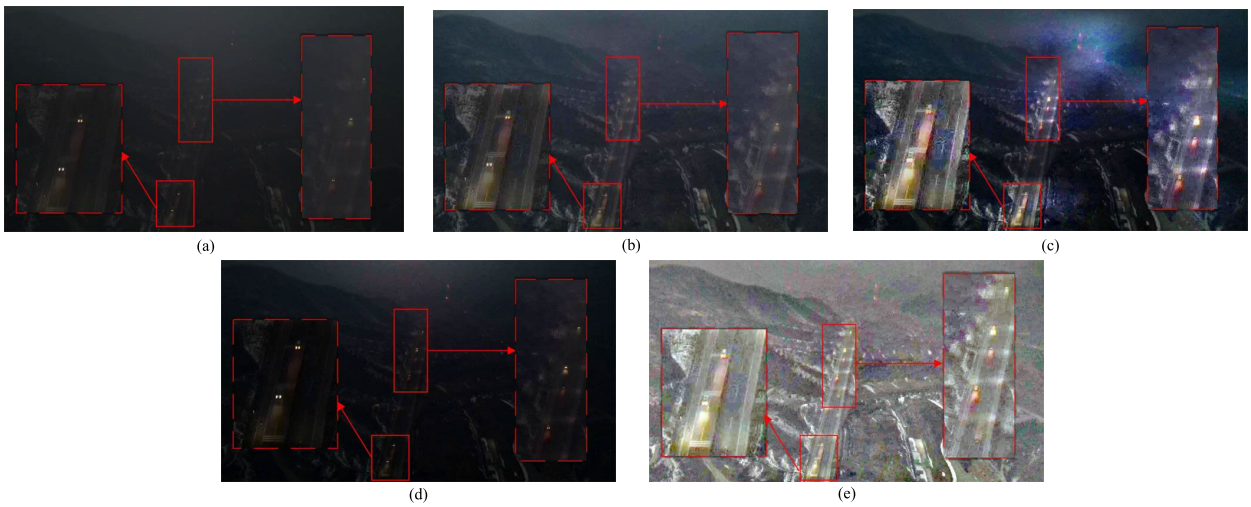

Fig. 12. Defogging effect of UAV images under low illumination with localized lighting: (a) original UAV image with fog, (b) algorithms in [5], (c) algorithms in [51], (d) algorithms in [54], and (e) algorithms in this article.

enhances the overall efficacy of perception algorithms in foggy images. The algorithm incorporating the perception-oriented feedback dehaze mechanism shows a comparative increase of 13.79% in mAP over the algorithm without it, even without the two filtering improvement algorithms. The objective dehazing evaluation metrics such as Γ, $\mathcal{G}$, K, and Λ demonstrate that the proposed filtering improvement algorithms substantially enhance dehazing for UAV images and maintain acceptable dehazing capabilities even without additional module integration, thanks to the SPSP effect. Regarding time consumption, our improved algorithms significantly enhance speed and reduce computational complexity. In the test environment, the proposed method requires only about 47 ms to dehaze a single image at 1920 $\times$ 1080 pixels resolution, meeting the real-time demands of a UAV monitoring system with 20 frames/s.

V. CONCLUSION

To address the challenges associated with eliminating haze in UAV imagery, a critical impediment in intelligent UAV applications, traditional methods often introduce problems such as color inconsistency, abnormal white areas, and perceptual distortions. To counter these issues, we propose a novel and highly effective concept termed "SPSP." This method adeptly identifies regions affected by haze and has been rigorously validated through extensive statistical and theoretical analyses. Utilizing this prior knowledge, we have devised a comprehensive dehazing framework for UAV images. This framework is designed to correct nonuniform illumination and eradicate haze under varied meteorological conditions within the Lab color space. Specifically, we have enhanced the homogeneous filtering algorithm and optimized the guided filtering process with SLIC, effectively addressing nonuniform illumination and producing superior dehazed images. Moreover, we introduce a co-evolutionary mechanism that synergizes haze removal and perceptual analysis, simultaneously processing super-pixel texture features and target detection outcomes while iteratively refining essential dehazing parameters through feedback loops. This innovative method not only improves the restoration of diverse target textures but also mitigates issues like contrast discrepancies and target blur during dehazing. The efficacy of our approach in UAV haze removal has been corroborated through a series of comprehensive experiments and assessments, encompassing subjective, objective, and perceptual aspects.

Although the proposed dehazing technique generally yields satisfactory outcomes for most UAV mid-altitude scenarios, it exhibits some limitations. In low-light conditions,

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the method may induce color anomalies in specific regions, as illustrated in Fig. 12. These limitations can be ameliorated using patch-based methods that refine local atmospheric light estimation and diminish visual artifacts. The challenge of UAV image dehazing remains an active area of research that necessitates further investigation.

In future studies, we plan to address these limitations and continue to enhance the efficacy of downstream tasks, optimize processing speeds, and improve visual results. The methodology introduced in this manuscript contributes significantly to the development of a priori knowledge-based dehazing theories and techniques. It effectively addresses common dehazing issues, such as the "white zone" phenomenon. Our forthcoming research will explore the utilization of the proposed a priori knowledge-based approach to develop an advanced UAV fog model for generating realistic synthetic datasets. In addition, we will investigate a comprehensive dehazing framework aimed at recovering high-quality scene radiance from UAV images across various weather conditions, times of day, and extreme weather scenarios.

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